

1. Consider the function $f(x, y) := |x| + 2|y|$.

(a) Find all the numbers k such that the level set $L_k := \{(x, y) \in \mathbb{R}^2: f(x, y) = k\}$ is not an empty set.

$$k \geq 0$$

(b) Sketch the sets $L_k \neq \emptyset$.

Each L_k with $k > 0$ is a rhombus with intercepts $(\pm k, 0)$ and $(0, \pm k/2)$ while $L_0 = \{(0, 0)\}$.

2. Evaluate the limit or show that it does not exist:

$$(i) \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x \sin y}{y \sin x} = \lim_{(x,y) \rightarrow (0,0)} \left[\frac{x}{\sin x} \right] \left[\frac{\sin y}{y} \right] = \lim_{(x,y) \rightarrow (0,0)} \left[\frac{x}{\sin x} \right] \times \lim_{(x,y) \rightarrow (0,0)} \left[\frac{\sin y}{y} \right] = 1;$$

(ii)

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} &= \lim_{(x,y) \rightarrow (0,0)} \frac{x(x - y)}{\sqrt{x} - \sqrt{y}} = \lim_{(x,y) \rightarrow (0,0)} \frac{x(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})}{\sqrt{x} - \sqrt{y}} \\ &= \lim_{(x,y) \rightarrow (0,0)} [x(\sqrt{x} + \sqrt{y})] = 0; \end{aligned}$$

(iii) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y}{y^2 + x}$. Find the limit along the line $y = x$, along the x -axis and conclude.

$$3. \text{ Consider the function } f(x, y) := \begin{cases} \frac{\sin^3(x)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

(a) Study the continuity of f at every $(x, y) \in \mathbb{R}^2$.

f is trivially continuous at every $(x, y) \neq (0, 0)$. Let us now study the continuity at $(0, 0)$.

Using $|\sin x| \leq |x|$ we get

$$0 \leq |f(x, y)| = \frac{|\sin x|^3}{x^2 + y^2} \leq \frac{|x|^3}{x^2 + y^2} = |x| \frac{x^2}{x^2 + y^2} \leq |x|.$$

Thus, by the squeeze theorem, it follows that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$. Hence, f is continuous at $(0, 0)$.

One could also use polar coordinates r, θ with $x = r \cos \theta$ and $y = r \sin \theta$. To this end, we first notice that along the x axis ($y = 0$) we get

$$f(x, 0) = \frac{\sin^3 x}{x^2} = \left[\frac{\sin x}{x} \right]^2 \times \sin x$$

and hence, $\lim_{x \rightarrow 0} f(x, 0) = \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]^2 \times \lim_{x \rightarrow 0} \sin x = 1 \times 0 = 0$.

Along the y -axis ($x = 0$), we have $\lim_{y \rightarrow 0} f(0, y) = \lim_{y \rightarrow 0} 0 = 0$. Therefore, we can consider $L = 0$ as the candidate limit of $f(x, y)$ as $(x, y) \rightarrow 0$. Now, we prove that $L = 0$ is indeed the limit using the polar coordinates as follows: We find through our knowledge of the properties of trig functions, a function $g(r)$ with $\lim_{r \rightarrow 0} g(r)$ such that

$$|f(r \cos \theta, r \sin \theta) - L| \leq g(r).$$

In that we we will say that $\lim_{r \rightarrow 0} f(r \cos \theta, r \sin \theta) = 0$ uniformly in θ .

Notice that

$$|f(r \cos \theta, r \sin \theta) - 0| = \frac{|\sin^3(r \cos \theta)|}{r^2} \leq \frac{|r^3 \cos \theta|^3}{r^2} = r |\cos \theta|^3 \leq r = g(r),$$

using $|\sin t| \leq |t|$ and $|\cos \theta| \leq 1$. So, it follows by the squeeze theorem that

$$\lim_{r \rightarrow 0} f(r \cos \theta, r \sin \theta) = 0 \text{ (uniformly in } \theta \text{)}.$$

Therefore, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$, and thus the function f is continuous at $(0, 0)$.

(b) Find (if they exist) $f_x(0, 0)$ and $f_y(0, 0)$.

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{\sin^3 x}{x^3} = 1$$

and

$$\lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0}{y} = 0.$$

Therefore, $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$.

4. Sketch the level curves and the graphs of the following functions:

(a) $f(x, y) = \frac{1}{x^2 + y^2};$

Let $k \in \mathbb{R}$. We want to sketch the set

$$L_k := \left\{ (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\} : \frac{1}{x^2 + y^2} = k \right\}$$

If $k \leq 0$, then $L_k = \emptyset$ because the function f is strictly positive, i.e., $f(x, y) > 0$ for every $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

If $k > 0$, then

$$L_k = \left\{ (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\} : x^2 + y^2 = \frac{1}{k} \right\}$$

and the level curves are a family of circles centered at the origin $(0, 0)$ with radius $1/\sqrt{k}$.

Now, to obtain the graph of f , we move each circle to its corresponding level (noticing that the smaller is the circle, the higher is its corresponding level) and then get the shape of the graph.

(b) $f(x, y) = x^2 + 3y^2$;

Let $k \in \mathbb{R}$. We want to sketch the set

$$L_k := \{(x, y) \in \mathbb{R}^2: x^2 + 3y^2 = k\}$$

If $k < 0$, then $L_k = \emptyset$ because the function f is positive, i.e., $f(x, y) \geq 0$ for every $(x, y) \in \mathbb{R}^2$.

If $k = 0$, then $L_0 = \{(0, 0)\}$.

If $k > 0$, then

$$L_k = \{(x, y) \in \mathbb{R}^2: x^2 + 3y^2 = k\} = \left\{ (x, y) \in \mathbb{R}^2: \frac{x^2}{k} + \frac{y^2}{\frac{k}{3}} \right\}.$$

and the level curves are a family of ellipses.

Now, to obtain the graph of f , we move each ellipse to its corresponding level (noticing that the larger is the ellipse, the higher is its corresponding level), and then get the shape of the graph which is called *elliptical paraboloid*

(c) $f(x, y) = e^{-x^2-y^2}$.

Let $k \in \mathbb{R}$. We want to sketch the set

$$L_k := \{(x, y) \in \mathbb{R}^2: e^{-x^2-y^2} = k\}$$

If $k \leq 0$, then $L_k = \emptyset$ because the function f is strictly positive, i.e., $f(x, y) > 0$ for every $(x, y) \in \mathbb{R}^2$. Also notice that $0 < f(x, y) \leq 1$.

If $k > 1$, then $L_k = \emptyset$.

If $0 < k < 1$, then

$$L_k = \{(x, y) \in \mathbb{R}^2: x^2 + y^2 = -\ln k\}$$

and the level curves are a family of circles centered at the origin $(0, 0)$ with radius $\sqrt{-\ln k}$.

If $k = 1$, then $L_1 = \{(0, 0)\}$.

Now, to obtain the graph of f , we move each circle to its corresponding level (noticing that the smaller is the circle, the higher is its corresponding level), and then get the shape of the graph.

5. Given the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = \frac{x+y}{2x^2+2y^2+4}$ for every $(x, y) \in \mathbb{R}^2$.

(i) Describe the level curves of the function f .

Let $k \in \mathbb{R}$. We want to sketch the sets

$$L_k := \left\{ (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}: \frac{x+y}{2x^2+2y^2+4} = k \right\}.$$

We notice that

$$\frac{x+y}{2x^2+2y^2+4} = k \Leftrightarrow 2k(x^2+y^2+2) = x+y.$$

If $k = 0$, then we get that $L_0 = \{(x, y) \in \mathbb{R}^2: x + y = 0\}$

If $k \neq 0$, then we get

$$\begin{aligned} x^2 + y^2 + 2 = \frac{x}{2k} + \frac{y}{2k} &\Leftrightarrow \left(x^2 - \frac{x}{2k}\right) + \left(y^2 - \frac{y}{2k}\right) = -2 \\ &\Leftrightarrow \left(x - \frac{1}{4k}\right)^2 + \left(y - \frac{1}{4k}\right)^2 = -2 + \frac{1}{8k^2} \\ &\Leftrightarrow \left(x - \frac{1}{4k}\right)^2 + \left(y - \frac{1}{4k}\right)^2 = \frac{1 - 16k^2}{8k^2} \quad (*) \end{aligned}$$

The rest is now according to whether the right-hand side of $(*)$ is positive or negative.

If $1 - 16k^2 < 0$ that is $k \in (-\infty, -1/4) \cup (1/4, \infty)$, then $L_k = \emptyset$ because the right-hand side of $(*)$ is strictly negative while the left-hand side is positive.

If $1 - 16k^2 > 0$ with $k \neq 0$, that is, $k \in (-1/4, 0) \cup (0, 1/4)$ then L_k is the circle centered at the point $(1/4k, 1/4k)$ with radius $\sqrt{\frac{1 - 16k^2}{8k^2}}$.

If $1 - 16k^2 = 0$, that is, $k = \pm 1/4$ then we get $L_{-1/4} = \{(-1, -1)\}$ and $L_{1/4} = \{(1, 1)\}$.

(ii) Find the level curve passing through the point $(0, 1)$.

Here we seek k such that $f(0, 1) = k$ and find that $k = f(0, 1) = \frac{1}{6}$ and we get that $L_{\frac{1}{6}}$ is the circle centered at $(3/2, 3/2)$ with radius $\sqrt{\frac{5}{2}}$.

(iii) Find both partial derivatives $f_x(1, -1)$ and $f_y(1, -1)$ of the function f at the point $(1, -1)$.

$$\begin{cases} f_x(x, y) = \frac{(2x^2 + 2y^2 + 4) - 4x(x + y)}{(2x^2 + 2y^2 + 4)^2}, \\ f_y(x, y) = \frac{(2x^2 + 2y^2 + 4) - 4y(x + y)}{(2x^2 + 2y^2 + 4)^2}, \end{cases} \Rightarrow f_x(1, -1) = f_y(1, -1) = \frac{1}{8}.$$

(iv) Write down the cartesian equation of the tangent plane to the graph of f at the point $(1, -1)$.

The tangent plane to the graph of f at the point $(1, -1)$ has cartesian equation

$$\begin{aligned} z = f(1, -1) + f_x(1, -1)(x - 1) + f_y(1, -1)(y + 1) &\Leftrightarrow z = 0 + \frac{1}{8}(x - 1) + \frac{1}{8}(y + 1) \\ &\Leftrightarrow \boxed{z = \frac{1}{8}x + \frac{1}{8}y}. \end{aligned}$$